This analysis examines the weekly stock performance of Apple Inc. and Meta Platforms Inc. over a one-year period using 53 weekly observations. The dataset includes the calculation of weekly returns, descriptive statistics, the relationship between the assets, and regression analysis with the market index. These statistical methods allow the evaluation of the behavior of the two stocks, their level of risk, and the extent to which their returns are influenced by overall market movements.

The summary statistics describe the main characteristics of the weekly returns. Apple has an average weekly return of approximately 0.0025, while Meta's mean return is slightly lower at about 0.0019. The standard deviation, which measures volatility and therefore risk, equals 0.0455 for Apple and 0.0503 for Meta. This indicates that Meta's returns fluctuate more widely than Apple's returns. In addition, the calculated variances are approximately 0.00207 for Apple and 0.00253 for Meta, confirming that Meta is the more volatile stock in this comparison. The minimum and maximum returns observed in the data also illustrate that both assets experience significant weekly swings, highlighting the uncertainty associated with stock market investments.

The relationship between the two assets was evaluated using covariance and correlation. The calculated correlation coefficient is about 0.33, indicating a moderate positive relationship between Apple and Meta returns. This means that the two stocks tend to move in the same general direction, although the connection is not very strong. Because the correlation is far from perfect, combining these two assets in a portfolio could provide diversification benefits and potentially reduce overall investment risk.

Regression analysis was conducted to determine how each company's return responds to changes in the market return. The market return was used as the independent variable, while the individual stock returns were treated as dependent variables. For Apple, the regression results show a beta coefficient of approximately 1.39, meaning that Apple's return tends to move more strongly than the market. A 1% change in market return is associated with an estimated 1.39% change in Apple's return. The R^2 value is around 0.49, which indicates that roughly 49% of the variation in Apple's weekly returns can be explained by movements in the market index. For Meta, the regression results indicate an even stronger response to market changes. The beta coefficient is approximately 1.46, suggesting that Meta's stock is slightly more sensitive to market fluctuations than Apple's stock. In this case, a 1% change in the market return is associated with an approximate 1.46% change in Meta's return. The R^2 value is about 0.44, meaning that around 44% of the variation in Meta's returns is explained by changes in the overall market.

A chi-square test was conducted to evaluate whether the population variance of the stock returns differs from the hypothesized variance. The test used 53 observations, giving 52 degrees of freedom, and the calculation was based on the sample variance obtained from the weekly returns. I used a chi-square test to check whether the annual standard deviation of the stock returns differs from 0.25 (25%), based on the sample variance of weekly returns.
An F-test was also performed to compare the variances of Apple and Meta in order to determine whether their levels of volatility differ. The test used the calculated sample variances of 0.00207 for Apple and 0.00253 for Meta. The comparison shows that Meta has a larger variance, indicating higher variability in weekly returns compared to Apple.

Overall, the statistical analysis shows that both Apple and Meta behave as relatively high-beta technology stocks that react strongly to market movements. Apple demonstrates slightly higher average returns and somewhat lower volatility, while Meta shows greater variability in returns. The moderate correlation between the two stocks indicates that combining them in a portfolio may still provide diversification benefits. These results highlight the importance of analyzing both return and risk when evaluating potential stock investments.